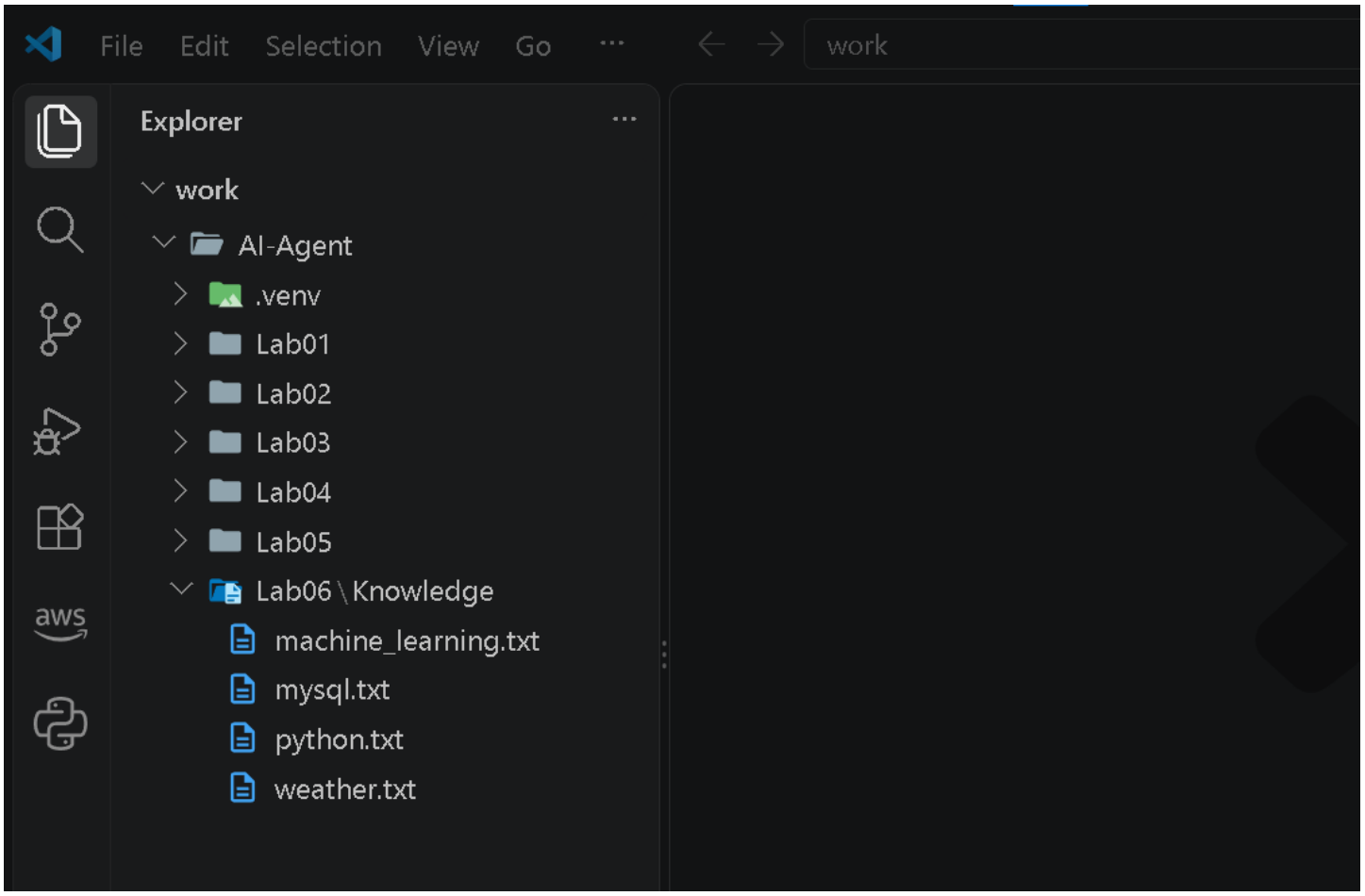
Build your First RAG System

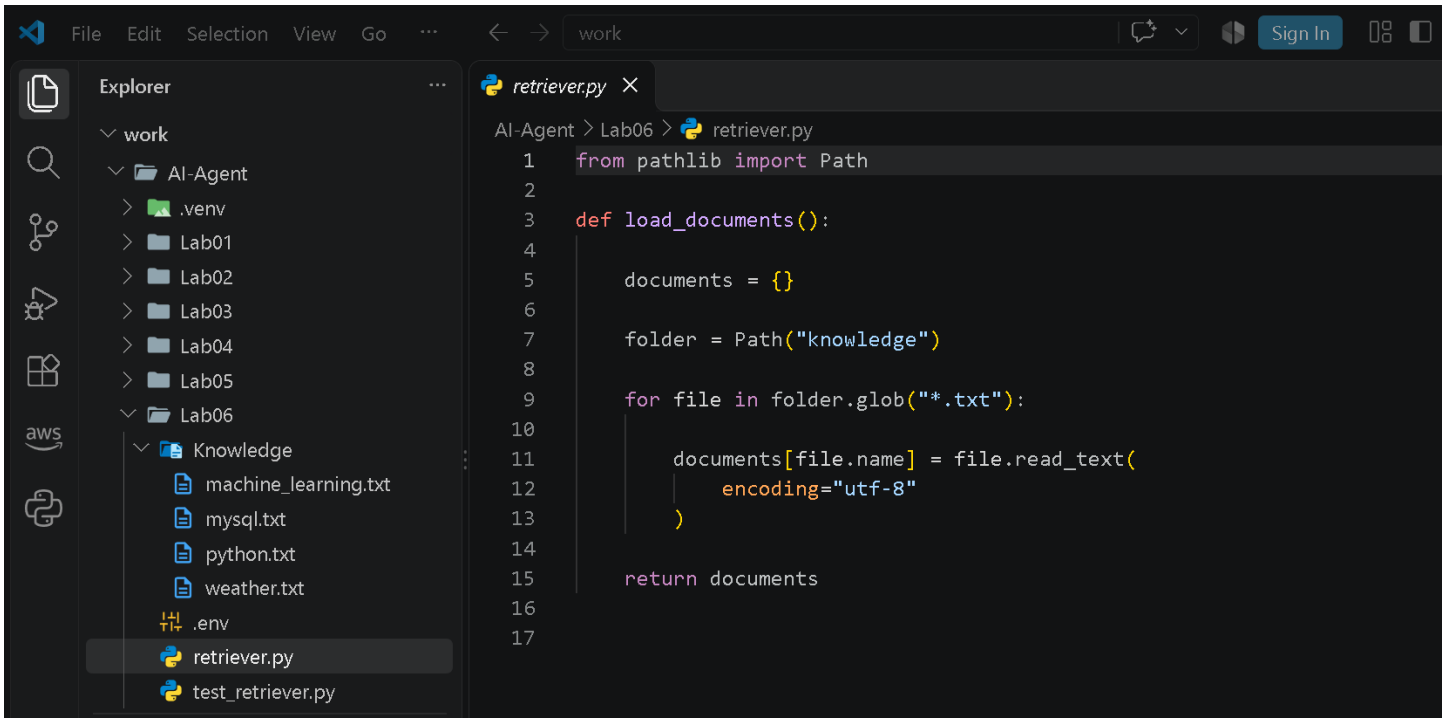
Summary: -

We built a RAG (Retrieval-Augmented Generation) system using Python, Ollama, and embedding models to retrieve relevant information from knowledge files and generate AI responses based on the provided data.

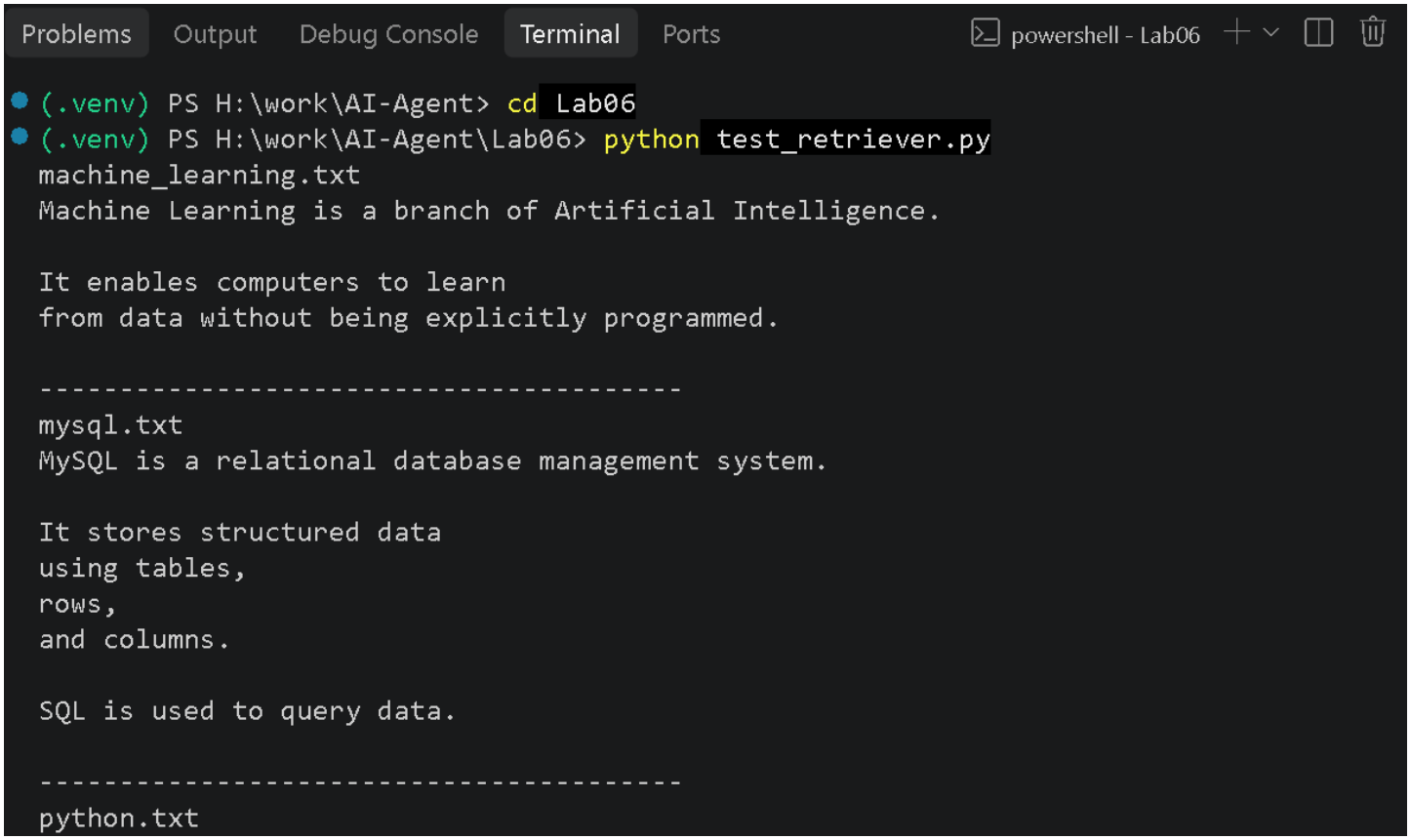
- Now Let's create a complete rag system.



- Let's create a folder Lab06 where we work on RAG System.
- In Lab06 create Folder Knowledge and inside this Folder add these 4 files provided in resource.



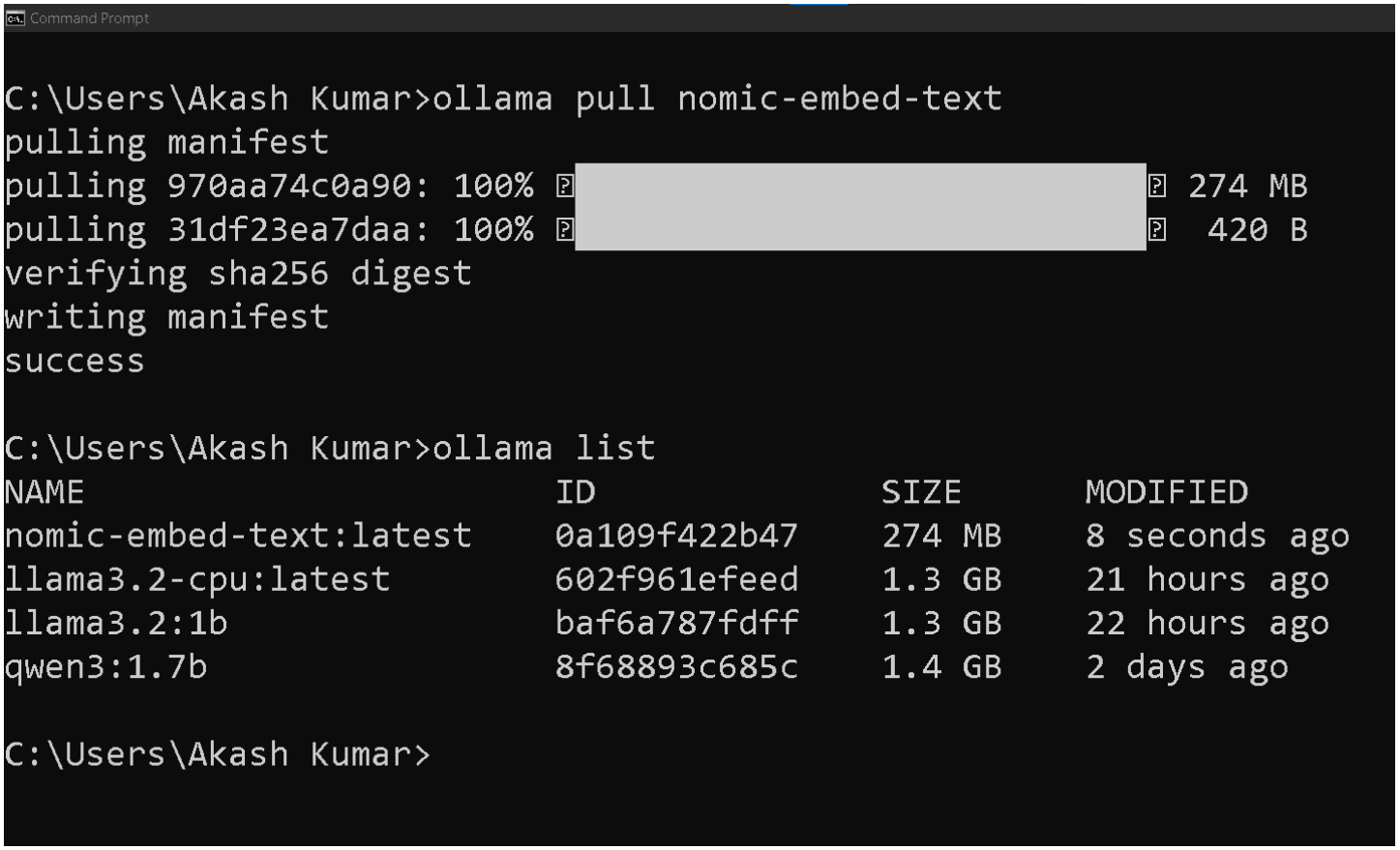
- Now Let's create two files retriever.py and test_retriever.py inside Lab06.
- In retriever.py we use our Knowledge files we can get input according to our Knowledge base.



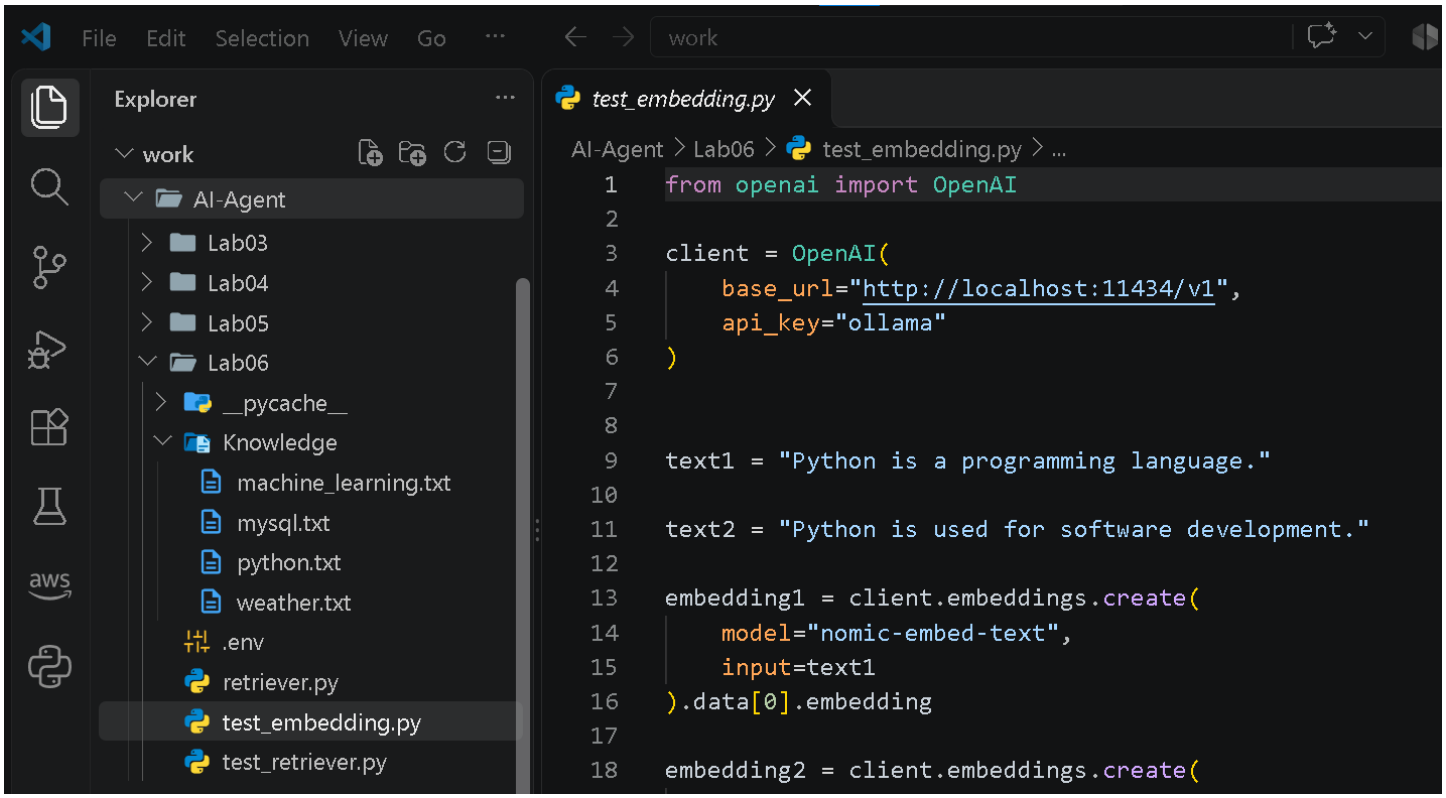
- See it's working.
- Now Let's install the embedding model.
- Go to your Command prompt.



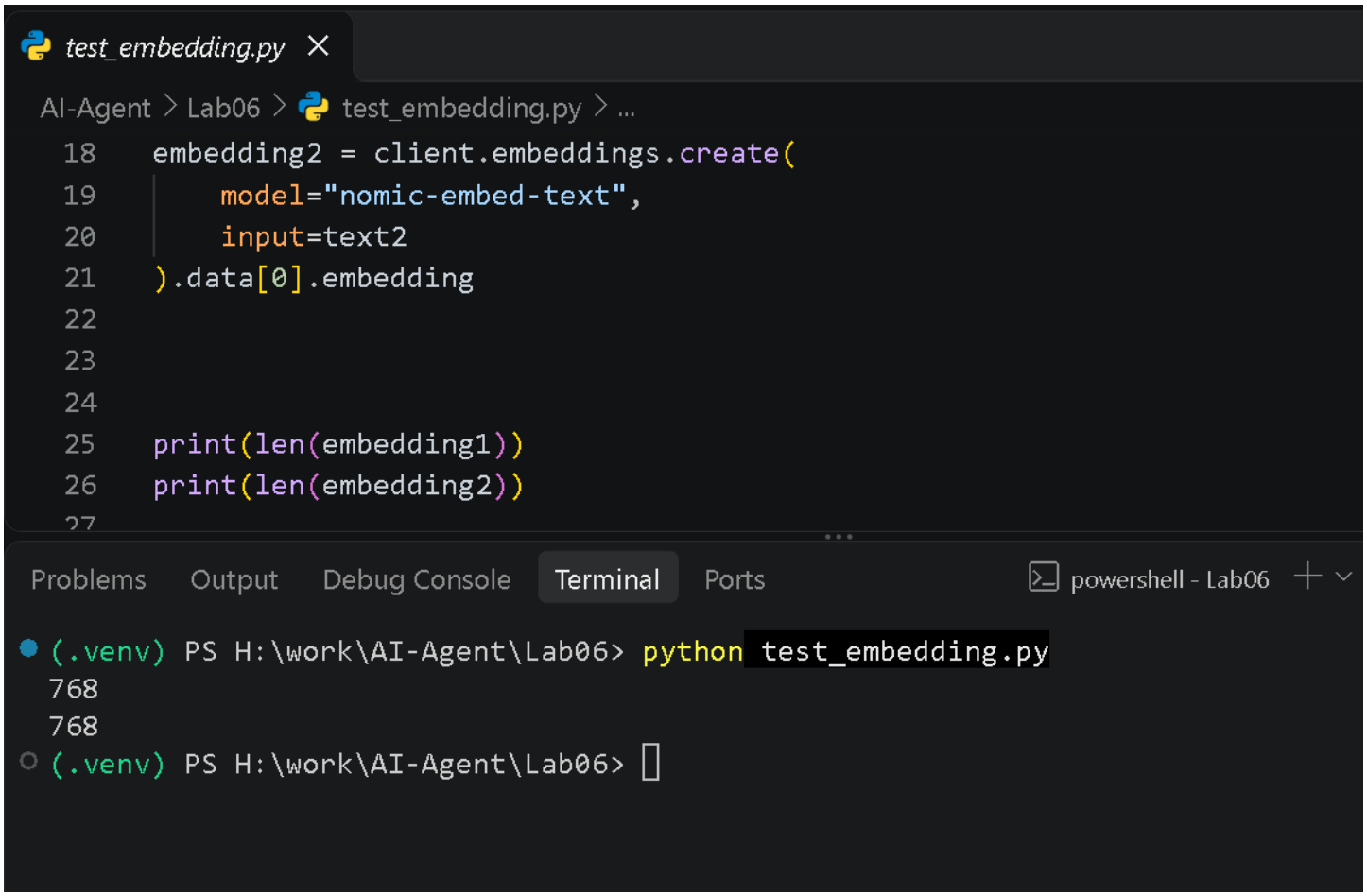
- Now In Command prompt type ollama pull nomic-embed-text
- After installation check its install or not.



- For this Enter ollama list
- Now see its install.
- Now go back to vs code

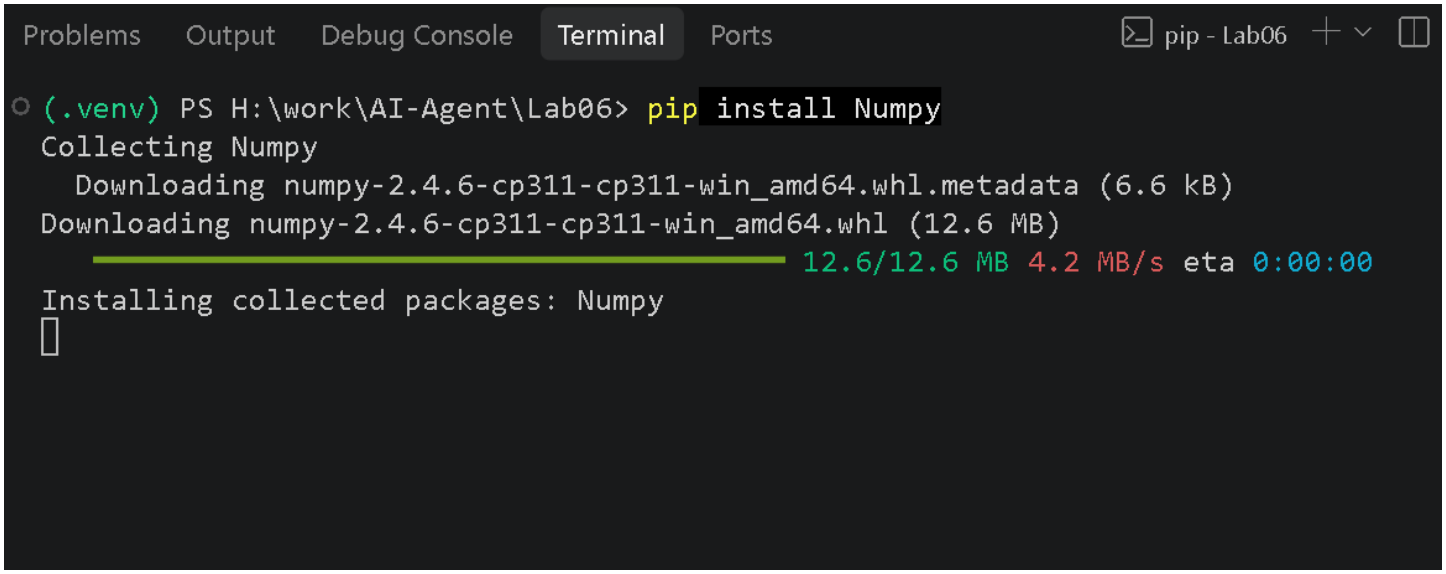


- Let's create one file inside lab06
- In this basically we check our embedding work correctly or not.
- Here we take two text text1 and text2 this model give length of Embedding.
- Let's see output.

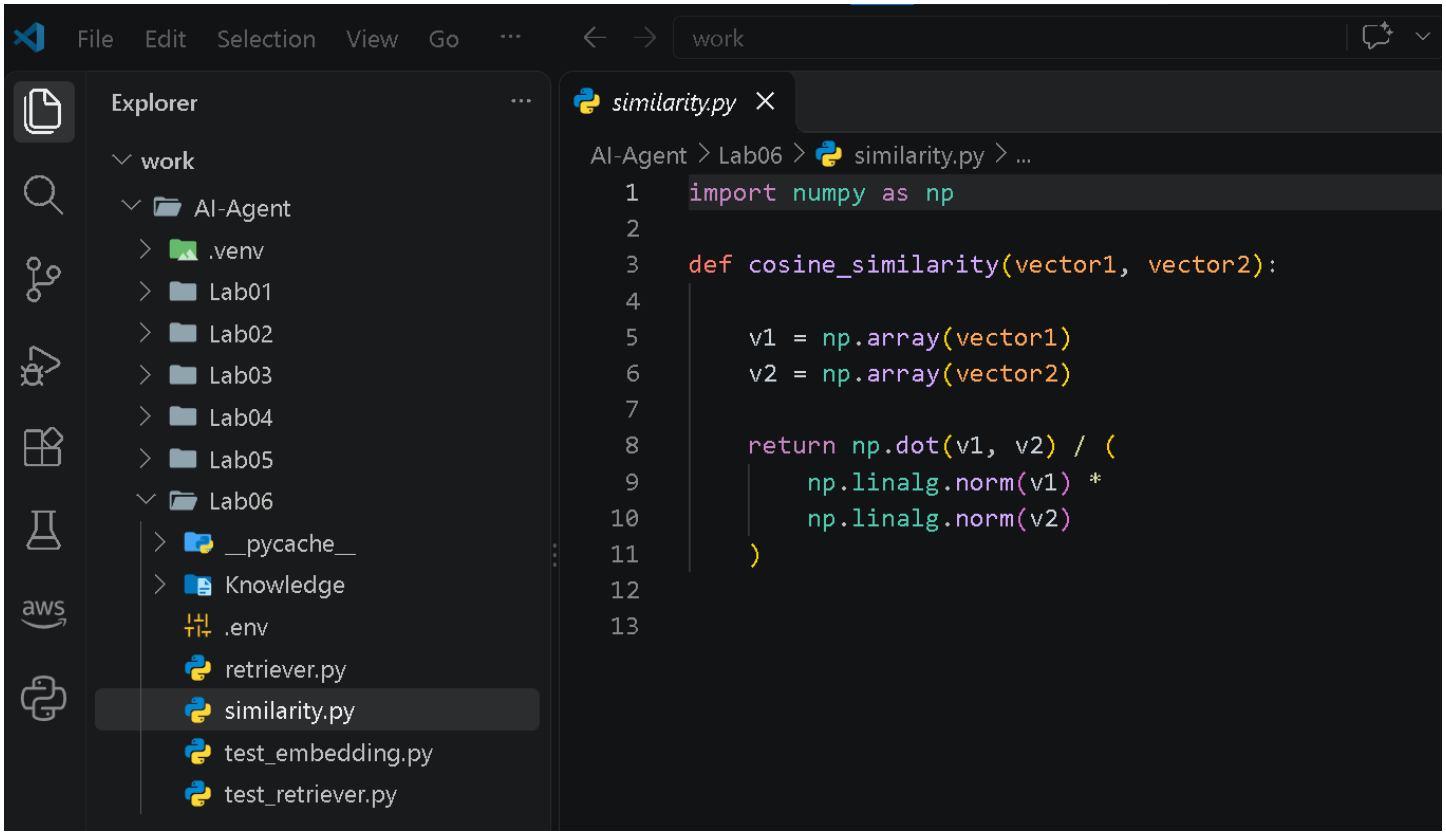


- Now see its give output and both are same but your Know text are different.
- So basically this is the length of vector and it doesn't matter your text is different it same.

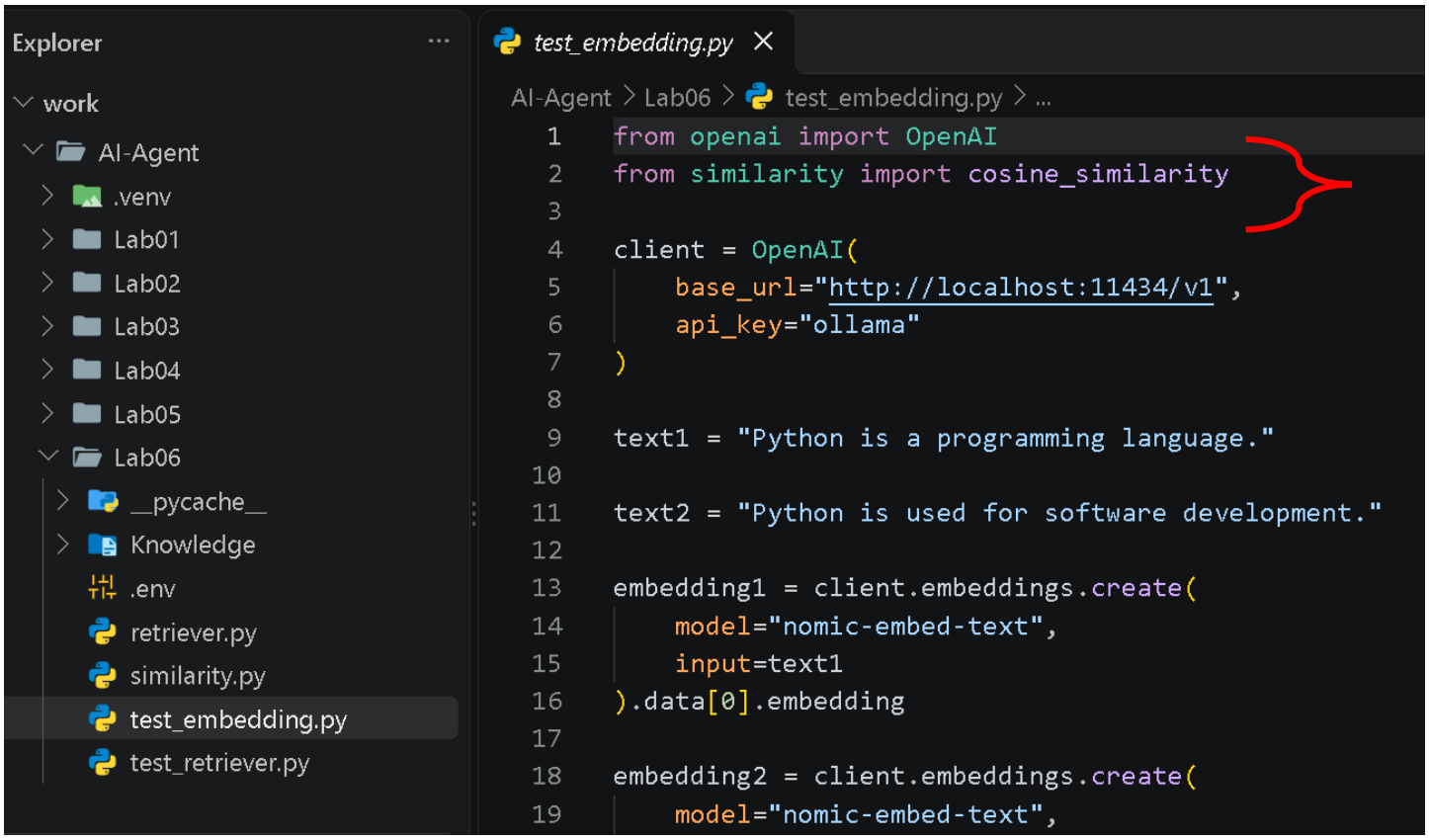
- Now suppose we compare a question with three documents.
- Every document has similarity score which one is higher it will retrieve data from it.
- This technique is called cosine similarity
- For our purpose remember this score
  - 1.00 Almost Identical
  - 0.90 very similar
  - 0.70 Fairly Similar
  - 0.30 weakly related
  - 0.0 unrelated
- We don't need to remember these score python library remember for us.



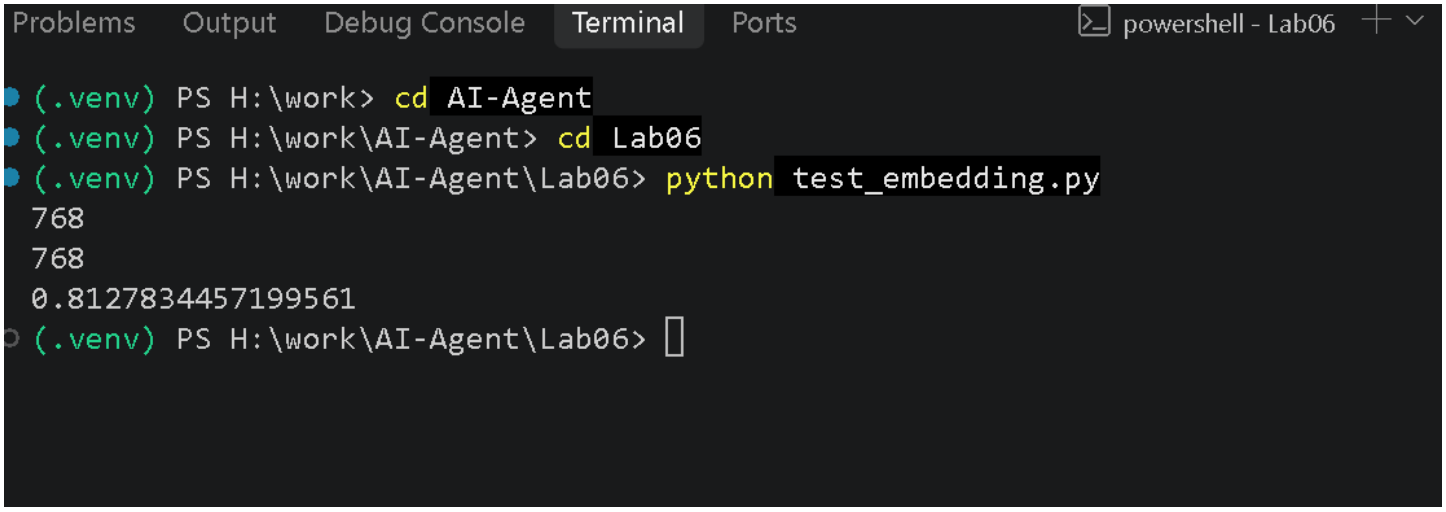
- Now go to Vs code open terminal and type pip install Numpy



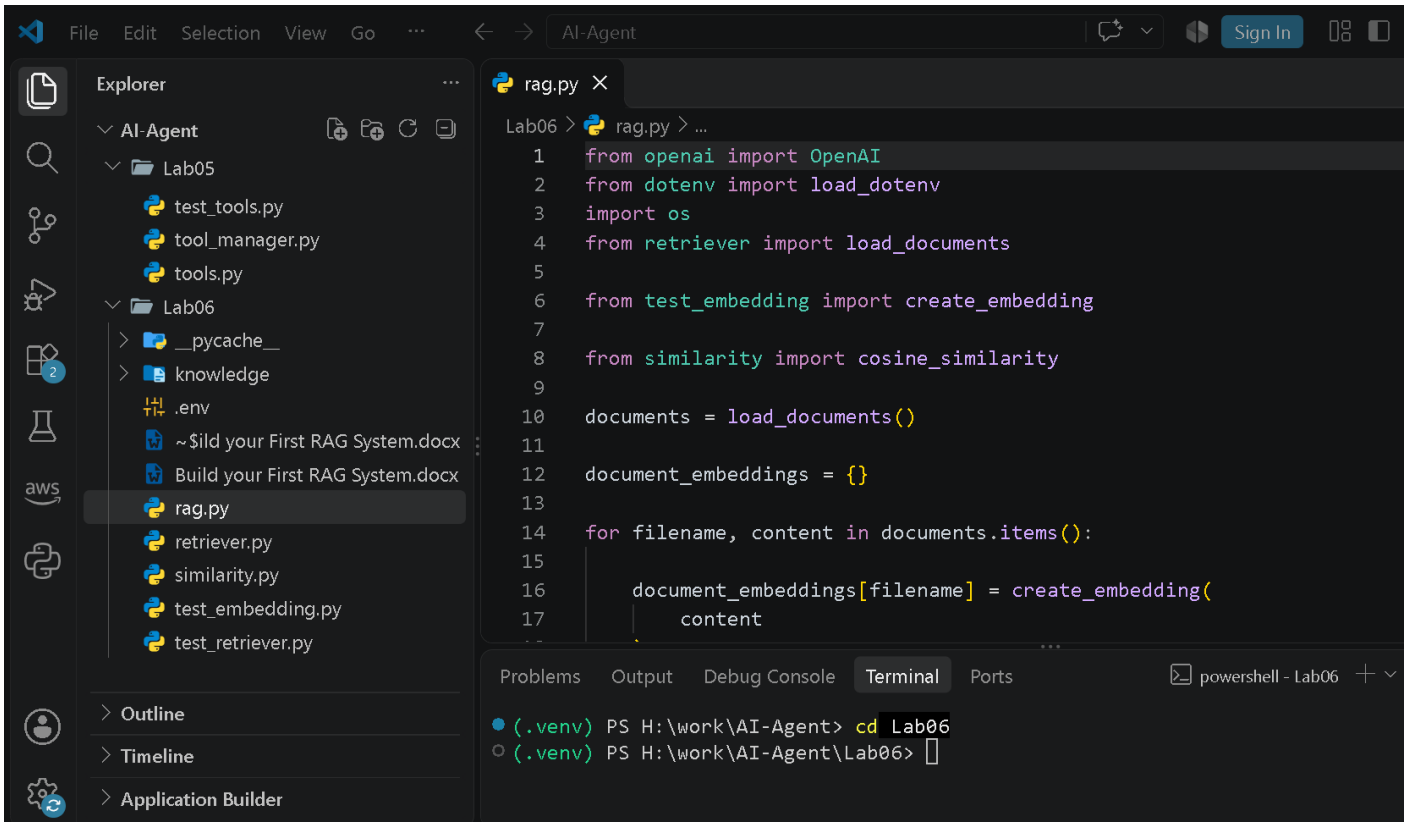
- Now Let's create a file similarity.py



- And import in test_embedding.py
- Now lets run this



- Now see it gives the similarity score.
- Now we add all these code in our Agent.



- Now lets run our rag.py

